

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	29-SEPTEMBER-2026
Team ID	6ab4c24ba238dc999eb5437a
Project Name	AI BlogNest API
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Register a user through the REST API using name, email and password.
FR-2	User Login & Authentication	Authenticate using email and password and issue JWT-based access.
FR-3	User Profile	Retrieve the authenticated user profile using a protected endpoint.
FR-4	Blog Management	Create, read, update and delete blog posts with title, content and category.
FR-5	AI Blog Generation	Generate blog content from a supplied topic using Gemini AI.
FR-6	AI Summarization	Generate a concise summary from supplied blog content using Gemini AI.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	REST endpoints use clear HTTP methods, JSON payloads and domain-based URL paths.
NFR-2	Security	Passwords are hashed with bcryptjs; protected endpoints use JWT bearer tokens and secrets are kept in environment variables.
NFR-3	Reliability	Central error middleware is configured and database connectivity is initialized when the server starts.
NFR-4	Performance	Express middleware and direct REST/database operations support lightweight API processing; no caching target is specified.
NFR-5	Availability	The application runs as a Node.js server and depends on MongoDB and Gemini services; distributed availability is not specified.
NFR-6	Scalability	Modular route separation for authentication, blogs and AI allows individual application areas to be extended independently